DESIGN THINKING & INNOVATION PROJECT

# GuardianPass

## Smart Hostel Gate Pass & Student Safety System

GuardianPass reimagines the hostel gate-pass process as a connected digital system — linking students, parents, wardens and security through real-time approvals, verified identity, and instant safety alerts.

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THE PROBLEM

# Life Without GuardianPass

*Manual, paper-based gate-pass systems leave gaps in safety, speed and communication.*

### Slow Manual Approval

Paper gate-passes depend on physically locating a warden, delaying every exit.

### Difficult Parent Verification

No reliable way to confirm a leave request genuinely comes from a parent.

### Manual Entry/Exit Records

Registers are hand-written, error-prone and hard to audit or search.

### Parents Kept in the Dark

Parents often don’t know when their child has actually left or returned.

### Delayed Emergency Response

No instant channel to alert parents or wardens when something goes wrong.

### Late-Return Blind Spot

Overdue students aren’t flagged automatically, so issues surface too late.

DISCOVER

# User Research & Empathy

## SURVEYS CONDUCTED WITH

- **Students**
- **Wardens**

## RESEARCH AREAS EXPLORED

- Gate-pass problems
- Parent communication gaps
- Entry / exit tracking
- Student safety concerns
- Emergency response needs
- Warden-side difficulties

## EMPATHY MAP

### SAYS

"Approval takes too long." "I never know if my child reached safely."

### THINKS

Wondering if the process is safe, fast and trustworthy for everyone involved.

### DOES

Fills paper forms, calls wardens, waits for manual sign-offs before leaving.

### FEELS

Anxious about delays, uncertain communication and safety while outside.

### PAINS

Slow approvals, unverifiable identity, no visibility once the student is out.

### NEEDS

Fast approval, verified consent, real-time updates and emergency support.

DEFINE

# The Design Challenge

HOW MIGHT WE…

*Make hostel students' travel safer while keeping parents and hostel authorities informed and connected?*

## KEY USER NEEDS TO ADDRESS

- **Fast Approval**
- **Genuine Parent Verification**
- **Accurate Records**
- **Better Communication**
- **Student Safety**
- **Emergency Support**

IDEATE

# Proposed Solution — GuardianPass




*A single connected platform links every stakeholder in the gate-pass journey.*

## CORE FEATURES

- Digital Gate Pass
- Parent App Approval
- IVR: 1=Approve, 0=Reject
- Warden Approval
- Secure QR Code
- QR Entry / Exit Scan
- Parent Notifications
- Approved-Trip Location
- SOS Emergency Alert
- Late-Return Alerts

HOW IT WORKS

# End-to-End Workflow

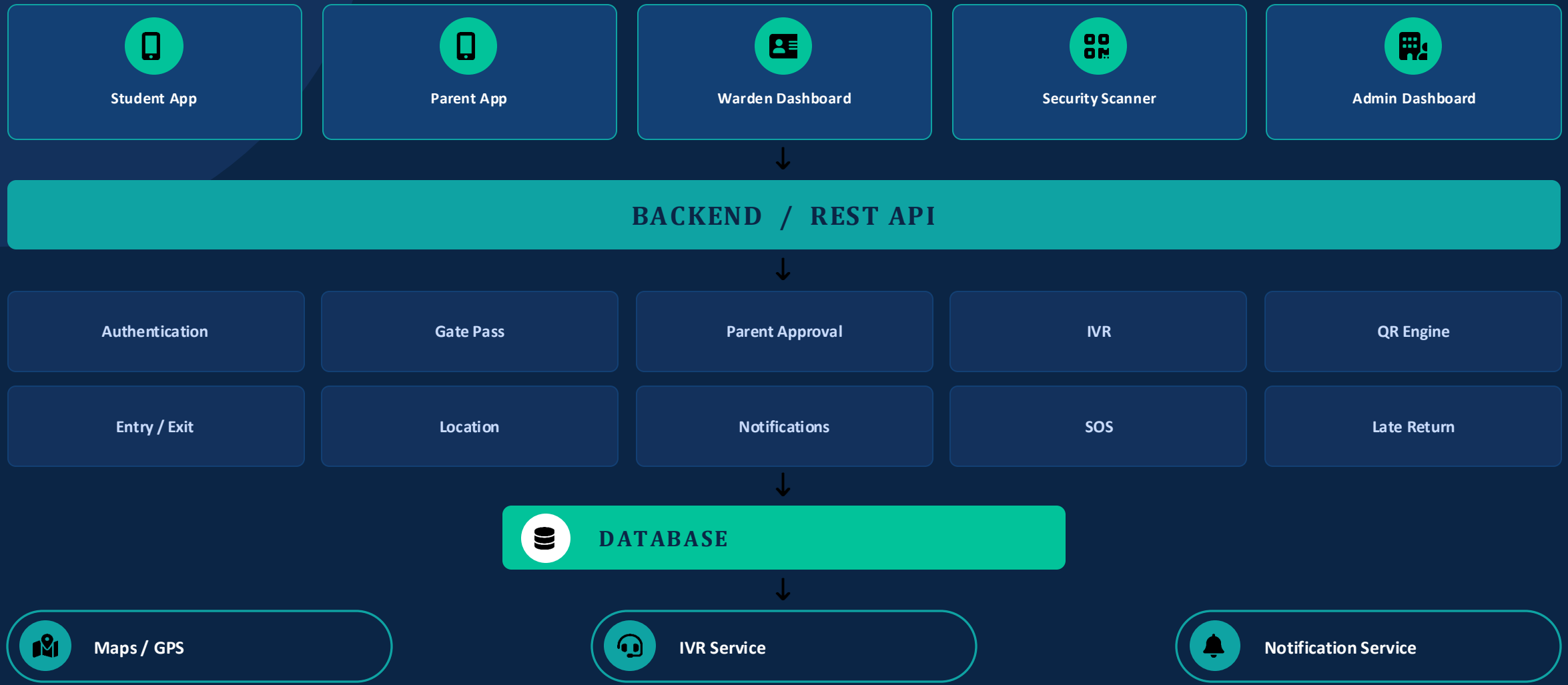
1. Student Applies
2. Parent Notified
3. App / IVR Approval
4. Warden Approval
5. QR Generated
6. Security Scans QR
7. Exit Recorded
8. Parent Notified
9. Active-Trip Location
10. Student Returns
11. QR Scanned
12. Entry + Late-Return Check

## ALTERNATE EMERGENCY FLOW

SOS Trigger → Instant Alert to Warden, Security & Parent → Live Location Shared → Immediate Response

TECHNICAL DESIGN

# System Architecture



BUILD

# Technology Stack

## Frontend

React / React Native

HTML, CSS, JavaScript

## Backend

Node.js, Express.js

REST API

## Database

PostgreSQL

MongoDB

## Security

JWT Authentication

Password Hashing

Role-Based Access

## Services

QR · GPS / Maps

IVR · Push Notifications

## Hosting

Cloud Server

Scalable Deployment

IMPACT

# Benefits, Innovation & Future Scope

## BENEFITS

- Safer students
- Parent involvement
- Faster approval
- Accurate records
- Better hostel monitoring
- Emergency communication
- Reduced paperwork

## INNOVATION

- App + IVR parent verification
- Two-level approval
- Secure QR gate pass
- Approved-trip location awareness
- SOS system
- Automatic late-return detection

## FUTURE SCOPE

- Geofencing
- AI safety analysis
- Smart CCTV integration
- Digital ID
- Face verification

CONCLUSION

# One Platform. Every Stakeholder.

GuardianPass connects students, parents, wardens and security on a single digital platform — replacing slow, manual gate-passes with fast approvals, verified consent, real-time visibility and instant emergency support.



Student



Parent



Warden



Security



**Safer Students**

+

**Connected Parents**

+

**Smarter Hostels**

## Thank You